

XMLParser

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Chapter 1

Hierarchical Index

1.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

CommandListener	7
String	8
Vector< T >	13
XmlAttribute	13
XmlFile	17
XmlObject	37
XmlComment	15
XmlNode	29
XmlText	39

Class Index

Here are the classes, structs, unions and interfaces with brief descriptions:

[illegible]

Chapter 3

File Index

3.1 File List

Here is a list of all documented files with brief descriptions:

Commands/ CommandListener.h	43
Containers/ Vector.hpp	43
String/ String.h	45
XML/ XmlAttribute.h	46
XML/ XmlComment.h	46
XML/ XmlFile.h	46
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Chapter 4

Class Documentation

4.1 CommandListener Class Reference

```
XML .  
#include <CommandListener.h>
```

Public Member Functions

- void [run](#) ([XmlFile](#) &file) .

4.1.1 Detailed Description

XML .

[XmlFile](#).

4.1.2 Member Function Documentation

4.1.2.1 [run\(\)](#)

```
void CommandListener::run (  
    XmlFile & file)
```

.

XML .

Parameters

file	XML , .
------	---------

The documentation for this class was generated from the following files:

- Commands/CommandListener.h
- Commands/CommandListener.cpp

4.2 String Class Reference

```
#include <String.h>
```

Public Member Functions

- `String ()`
- `String (const String &other)`
- `String (String &&other)`
- `String (const char *str)`
C-
- `String & operator= (const String &other)`
- `String & operator= (String &&other)`
- `~String ()`
- `char & operator[] (size_t ind)`
- `void append (const String &other)`
- `String & operator+= (const String &other)`
- `void clear ()`
- `const char * getPtr () const`
C-
- `const char & operator[] (size_t index) const`
- `operator bool () const`
- `void push_back (char c)`
- `size_t getLen () const`

Friends

- `std::ostream & operator<< (std::ostream &os, const String &other)`
- `bool operator< (String const &lhs, String const &rhs)`
- `String operator+ (const String &lhs, const String &rhs)`
- `std::istream & operator>> (std::istream &is, String &str)`
- `bool operator== (String const &lhs, String const &rhs)`

4.2.1 Detailed Description

String is a class that represents a string. It is a wrapper around the C++ `std::string` class. It provides a set of methods for manipulating strings, such as `length()`, `substr()`, and `find()`. It also provides a set of operators for string manipulation, such as `+`, `+`, and `+`.

4.2.2 Constructor & Destructor Documentation

4.2.2.1 String() [1/3]

String::String (
const [String](#) & other)

String::String ()

Parameters

other	,	.
-------	---	---

4.2.2.2 String() [2/3]

String::String (
[String](#) && other)

String::String ()

Parameters

other	,	.
-------	---	---

4.2.2.3 String() [3/3]

String::String (
const char * str)

String::String (C-
char * str)

Parameters

str	-	.
-----	---	---

4.2.3 Member Function Documentation

4.2.3.1 append()

```
void String::append (
    const String & other)
```

.

Parameters

other	.
-------	---

4.2.3.2 getLen()

```
size_t String::getLen () const
```

.

Returns

.

4.2.3.3 getPtr()

```
const char * String::getPtr () const
```

C- .

Returns

.

4.2.3.4 operator bool()

```
String::operator bool () const
```

.

Returns

```
    true .
```


4.2.3.5 operator+=()

`String` & `String::operator+=` (
const `String` & other)

.

Parameters

other	.
-------	---

Returns

.

4.2.3.6 operator=() [1/2]

`String` & `String::operator=` (
const `String` & other)

.

Parameters

other	,	.
-------	---	---

Returns

.

4.2.3.7 operator=() [2/2]

`String` & `String::operator=` (
`String` && other)

.

Parameters

other	,	.
-------	---	---

Returns

.

4.2.3.8 operator[]() [1/2]

```
char & String::operator[] (
    size_t ind)
```

Parameters

ind	.
-----	---

Returns

4.2.3.9 operator[]() [2/2]

```
const char & String::operator[] (
    size_t index) const
```

Parameters

index	.
-------	---

Returns

4.2.3.10 push_back()

```
void String::push_back (
    char c)
```

Parameters

c	.
---	---

4.2.4 Friends And Related Symbol Documentation

4.2.4.1 operator+

```
String operator+ (
    const String & lhs,
    const String & rhs) [friend]
    .
```

Returns

, .

The documentation for this class was generated from the following files:

- String/String.h
- String/String.cpp

4.3 Vector< T > Class Template Reference

Public Member Functions

- Iterator begin ()
- Iterator end ()
- Iterator begin () const
- Iterator end () const
- bool empty () const
- operator bool () const
- size_t getSize () const
- void print (std::ostream &os=std::cout) const
- void push_back (const T &el)
- T pop_back ()
- T & operator[] (size_t ind)
- const T & operator[] (size_t ind) const
- void clear ()
- Vector (Vector &&other)
- Vector (const Vector &other)
- Vector & operator= (const Vector &other)
- Vector & operator= (Vector &&other)

The documentation for this class was generated from the following file:

- Containers/Vector.hpp

4.4 XmlAttribute Class Reference

XML .

```
#include <XmlAttribute.h>
```

Public Member Functions

- `XmlAttribute ()`
XML .
- `XmlAttribute (const String &name, const String &value)`
XML .
- `void setValue (const String &val)` .
- `const String & getName () const` .
- `const String & getValue () const` .

4.4.1 Detailed Description

XML .

XML .

4.4.2 Constructor & Destructor Documentation

4.4.2.1 XmlAttribute()

`XmlAttribute::XmlAttribute (`
 `const String & name,`
 `const String & value)`

XML .

Parameters

name	.
value	.

4.4.3 Member Function Documentation

4.4.3.1 getName()

`const String & XmlAttribute::getName () const`

.

Returns

.

4.4.3.2 getValue()

```
const String & XmlAttribute::getValue () const
```

Returns

4.4.3.3 setValue()

```
void XmlAttribute::setValue (
    const String & val)
```

Parameters

val	.
-----	---

The documentation for this class was generated from the following files:

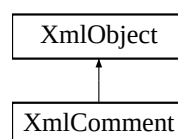
- XML/XmlAttribute.h
- XML/XmlAttribute.cpp

4.5 XmlComment Class Reference

```
XML      XML      .
```

```
#include <XmlComment.h>
```

Inheritance diagram for XmlComment:



Public Member Functions

- const String & getComment () const
- void serialize (std::ostream &os, size_t indent=0) const override
XML
- void printText (std::ostream &os=std::cout) const override
- XmlObject * clone () const override
- XmlComment (const String &str)
XML

Public Member Functions inherited from [XmlObject](#)

- virtual `~XmlObject ()=default`
- void [printIndents](#) (std::ostream &os, size_t indent=0) const

4.5.1 Detailed Description

XML XML .

XML

4.5.2 Constructor & Destructor Documentation

4.5.2.1 XmlComment()

XmlComment::XmlComment (
 const [String](#) & str)

XML .

Parameters

str	.
-----	---

4.5.3 Member Function Documentation

4.5.3.1 clone()

[XmlObject](#) * XmlComment::clone () const [override], [virtual]

Returns

Implements [XmlObject](#).

4.5.3.2 getComment()

const [String](#) & XmlComment::getComment () const

Returns

4.5.3.3 printText()

```
void XmlComment::printText (
    std::ostream & os = std::cout) const    [override], [virtual]
```

Parameters

os	.
----	---

Implements [XmlObject](#).

4.5.3.4 serialize()

```
void XmlComment::serialize (
    std::ostream & os,
    size_t indent = 0) const    [override], [virtual]
```

XML .

Parameters

os	.
indent	.

Implements [XmlObject](#).

The documentation for this class was generated from the following files:

- XML/XmlComment.h
- XML/XmlComment.cpp

4.6 XmlFile Class Reference

XML , , XML .

```
#include <XmlFile.h>
```

Public Member Functions

- `XmlFile ()`
- `XmlFile (const XmlFile &other)`
- `XmlFile (XmlFile &&other)`
- `XmlFile & operator= (const XmlFile &other)`
- `XmlFile & operator= (XmlFile &&other)`
- `~XmlFile ()`
- `void deserialize (std::istream &is)`
- `void serialize (std::ostream &os) const`
- `XmlNode * getById (String id)`
- `XmlComment * createComment (std::istream &is)`
- `XmlNode * createNode (std::istream &is)`
- `void addAttribute (std::istream &is, XmlNode &child)`
- `String createValue (std::istream &is)`
- `String createAttribute (std::istream &is)`
- `void setName (std::istream &is, XmlNode &child, char first)`
- `String toString (size_t num)`
- `void select (const String &id, const String &key) const`
- `void set (const String &id, const String &key, const String &value)`
- `void children (const String &id) const`
- `void child (const String &id, size_t pos) const`
- `void text (const String &id) const`
- `void deleteAttr (const String &id, const String &key)`
- `void newChild (const String &id)`
- `void open (const String &filePath)`
- `void close ()`

- void save () const
- void saveAs (const String &fileName) const
- const XmlNode & getByTagName (const String &tagName) const
- XmlText * createText (std::istream &is, char first)
- void reachEndOfTag (std::istream &is)
- bool isWhitespaceOnly (const String &str)
- void buildTree (std::istream &is, XmlNode &curNode)
- void handleId (XmlNode &child)
- const XmlNode * getId (const String &id) const
- Vector< XmlNode * > applyQuery (const Vector< XmlNode * > &curResult, const String &miniQuery) const
- Vector< String > splitBySlash (const String &query) const
- Vector< XmlNode * > filterInd (const Vector< XmlNode * > &nodes, const String &miniQuery, size_t filterStart) const
- Vector< XmlNode * > filterAllAttr (const Vector< XmlNode * > &nodes, const String &miniQuery, size_t filterStart) const
- Vector< XmlNode * > filterByText (const Vector< XmlNode * > &nodes, const String &miniQuery, size_t filterStart) const
- String getNameFromQuery (const String &query, size_t start=0) const
- Vector< XmlNode * > applyFilter (const Vector< XmlNode * > &nodes, const String &miniQuery, size_t filterStart) const
- void xpath (const String &query) const

4.6.1 Detailed Description

XML , XML .

XML , XPath-

4.6.2 Constructor & Destructor Documentation

4.6.2.1 XmlFile() [1/2]

XmlFile::XmlFile (
 const [XmlFile](#) & other)

.

Parameters

other	,	.
-------	---	---

4.6.2.2 XmlFile() [2/2]

XmlFile::XmlFile (
 [XmlFile](#) && other)

.

Parameters

other	,	.
-------	---	---

4.6.3 Member Function Documentation

4.6.3.1 addAttribute()

void XmlFile::addAttribute (
 std::istream & is,
 [XmlNode](#) & child)

.

Parameters

is	.
child	, .

4.6.3.2 buildTree()

```
void XmlFile::buildTree (  
    std::istream & is,  
    XmlNode & curNode)
```

XML .

Parameters

is	.
curNode	.

4.6.3.3 child()

```
void XmlFile::child (  
    const String & id,  
    size_t pos) const
```

.

Parameters

id	.
pos	.

4.6.3.4 children()

```
void XmlFile::children (  
    const String & id) const
```

.

Parameters

id	.
----	---

4.6.3.5 createAttribute()

```
String XmlFile::createAttribute (  
    std::istream & is)
```

.

Parameters

is	.
----	---

Returns

.

4.6.3.6 createComment()

```
XmlComment * XmlFile::createComment (
    std::istream & is)
```

XML .

Parameters

is	.
----	---

Returns

.

4.6.3.7 createNode()

```
XmlNode * XmlFile::createNode (
    std::istream & is)
```

XML .

Parameters

is	.
----	---

Returns

.

4.6.3.8 createText()

```
XmlText * XmlFile::createText (
    std::istream & is,
    char first)
```

.

Parameters

is	.
first	.

Returns

.

4.6.3.9 createValue()

```
String XmlFile::createValue (  
    std::istream & is)
```

.

Parameters

is	.
----	---

Returns

.

4.6.3.10 deleteAttr()

```
void XmlFile::deleteAttr (  
    const String & id,  
    const String & key)
```

.

Parameters

id	.
key	.

4.6.3.11 deserialize()

```
void XmlFile::deserialize (  
    std::istream & is)
```

XML .

Parameters

is	.
----	---

4.6.3.12 `getById()` [1/2]

```
const XmlNode * XmlFile::getById (  
    const String & id) const
```

.

Parameters

id	.
----	---

Returns

.

4.6.3.13 `getById()` [2/2]

```
XmlNode * XmlFile::getById (  
    String id)
```

.

Parameters

id	.
----	---

Returns

.

4.6.3.14 `getByTagName()`

```
const XmlNode & XmlFile::getByTagName (  
    const String & tagName) const
```

.

Parameters

tagName	.
---------	---

Returns

.

4.6.3.15 handleId()

```
void XmlFile::handleId (  
    XmlNode & child)
```

Parameters

child	.
-------	---

4.6.3.16 isWhitespaceOnly()

```
bool XmlFile::isWhitespaceOnly (  
    const String & str)
```

Parameters

str	.
-----	---

Returns

true .

4.6.3.17 newChild()

```
void XmlFile::newChild (  
    const String & id)
```

Parameters

id	.
----	---

4.6.3.18 open()

```
void XmlFile::open (  
    const String & filePath)
```

XML .

Parameters

filePath	.
----------	---

4.6.3.19 operator=() [1/2]

[XmlFile](#) & XmlFile::operator= (
 const [XmlFile](#) & other)

.

Parameters

other	,	.
-------	---	---

Returns

.

4.6.3.20 operator=() [2/2]

[XmlFile](#) & XmlFile::operator= (
 [XmlFile](#) && other)

.

Parameters

other	,	.
-------	---	---

Returns

.

4.6.3.21 reachEndOfTag()

void XmlFile::reachEndOfTag (
 std::istream & is)

.

Parameters

is	.
----	---

4.6.3.22 saveAs()

```
void XmlFile::saveAs (  
    const String & fileName) const
```

.

Parameters

fileName	.
----------	---

4.6.3.23 select()

```
void XmlFile::select (  
    const String & id,  
    const String & key) const
```

.

Parameters

id	.
key	.

4.6.3.24 serialize()

```
void XmlFile::serialize (  
    std::ostream & os) const
```

XML .

Parameters

os	.
----	---

4.6.3.25 set()

```
void XmlFile::set (  
    const String & id,  
    const String & key,  
    const String & value)
```

.

Parameters

id	.
----	---

key	.
value	.

4.6.3.26 setName()

```
void XmlFile::setName (
    std::istream & is,
    XmlNode & child,
    char first)
```

XML .

Parameters

is	.
child	, .
first	.

4.6.3.27 text()

```
void XmlFile::text (
    const String & id) const
```

.

Parameters

id	.
----	---

4.6.3.28 toString()

```
String XmlFile::toString (
    size_t num)
```

.

Parameters

num	.
-----	---

Returns

.

4.6.3.29 XPath()

```
void XmlFile::XPath (
    const String & query) const
```

XPath- .

Parameters

query	.
-------	---

The documentation for this class was generated from the following files:

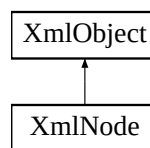
- XML/XmlFile.h
- XML/XmlFile.cpp

4.7 XmlNode Class Reference

XML XML .

```
#include <XmlNode.h>
```

Inheritance diagram for XmlNode:



Public Member Functions

- XmlNode ()
- XML .
- XmlNode (const XmlNode &other)
- XmlNode (XmlNode &&other)
- XmlNode & operator= (XmlNode &&other)
- XmlNode & operator= (const XmlNode &other)
- ~XmlNode ()
- XML .
- const String & getId () const
- void addAttribute (const XmlAttribute &attribute)
- void addChild (XmlObject *child)

- const [Vector](#)< [XmlAttribute](#) > & [getAttributes](#) () const
- [Vector](#)< [XmlAttribute](#) > & [getAttributes](#) ()
- void [printText](#) (std::ostream &os=std::cout) const override
- [String](#) [getTextContent](#) () const
- [Vector](#)< [XmlNode](#) * > [getChildrenByName](#) (const [String](#) &name) const
- int [getAttributeInd](#) (const [String](#) &attr) const
- [XmlNode](#) (const [String](#) &name)
- XML
const [String](#) & [getName](#) () const
- void [setName](#) (const [String](#) &name)
- const [Vector](#)< [XmlObject](#) * > & [getChildren](#) () const
- void [serialize](#) (std::ostream &os, size_t indent=0) const override
- XML
[XmlObject](#) * [clone](#) () const override

Public Member Functions inherited from [XmlObject](#)

- virtual ~[XmlObject](#) ()=default
- void [printIndents](#) (std::ostream &os, size_t indent=0) const

4.7.1 Detailed Description

XML XML .
 , , , ,

4.7.2 Constructor & Destructor Documentation

4.7.2.1 [XmlNode](#)() [1/3]

[XmlNode](#)::[XmlNode](#) (
 const [XmlNode](#) & other)

Parameters

other	, .
-------	-----

4.7.2.2 XmlNode() [2/3]

XmlNode::XmlNode (
[XmlNode](#) && other)

.

Parameters

other	, .
-------	-----

4.7.2.3 XmlNode() [3/3]

XmlNode::XmlNode (
const [String](#) & name)

XML .

Creates an XML node with a given name.

Parameters

name	XML .
name	Name of the XML element.

4.7.3 Member Function Documentation

4.7.3.1 addAttribute()

void XmlNode::addAttribute (
const [XmlAttribute](#) & attribute)

.

Adds an attribute to the XML node.

Parameters

attribute	.
attribute	Attribute to add.

4.7.3.2 addChild()

```
void XmlNode::addChild (
    XmlObject * child)
```

.

Adds a child XML object to the node.

Parameters

child	.
-------	---

The node takes ownership of the provided pointer.

Parameters

child	Child object to add.
-------	----------------------

4.7.3.3 clone()

```
XmlObject * XmlNode::clone () const  [override], [virtual]
```

.

Returns

.

Implements [XmlObject](#).

4.7.3.4 getAttributeInd()

```
int XmlNode::getAttributeInd (
    const String & attr) const
```

.

Finds the index of an attribute by name.

Parameters

attr	.
------	---

Returns

-1, .

Parameters

attr	Name of the attribute to search for.
------	--------------------------------------

Returns

Index of the attribute if found, otherwise -1.

4.7.3.5 getAttributes() [1/2]

`Vector< XmlNodeAttribute > & XmlNode::getAttributes ()`

.

Returns the node attributes.

Returns

Reference to the collection of attributes.

4.7.3.6 getAttributes() [2/2]

`const Vector< XmlNodeAttribute > & XmlNode::getAttributes () const`

.

Returns the node attributes.

Returns

Constant reference to the collection of attributes.

4.7.3.7 getChildren()

`const Vector< XmlNode * > & XmlNode::getChildren () const`

.

Returns the child XML objects of the node.

Returns

Constant reference to the collection of child objects.

4.7.3.8 getChildrenByName()

```
Vector< XmlNode * > XmlNode::getChildrenByName (
    const String & name) const
```

.

Finds all child elements with a given name.

Parameters

name	.
------	---

Returns

XML .

Parameters

name	Name of the child elements to search for.
------	---

Returns

Collection of matching child nodes.

4.7.3.9 getId()

```
const String & XmlNode::getId () const
```

.

Returns

.

4.7.3.10 getName()

```
const String & XmlNode::getName () const
```

.

Returns

.

4.7.3.11 getTextContent()

[String](#) XmlNode::getTextContent () const

Collects the direct text content of the XML node.

Returns

[String](#) containing the text from direct text children.

4.7.3.12 operator=() [1/2]

[XmlNode](#) & XmlNode::operator= (
const [XmlNode](#) & other)

Parameters

other	,	.
-------	---	---

Returns

4.7.3.13 operator=() [2/2]

[XmlNode](#) & XmlNode::operator= (
[XmlNode](#) && other)

Parameters

other	,	.
-------	---	---

Returns

4.7.3.14 printText()

```
void XmlNode::printText (
    std::ostream & os = std::cout) const    [override], [virtual]
```

.

Prints the text content of all child objects.

Parameters

os	.
os	Output stream to print to.

Implements [XmlNode](#).

4.7.3.15 serialize()

```
void XmlNode::serialize (
    std::ostream & os,
    size_t indent = 0) const    [override], [virtual]
```

XML

.

Writes the XML node and its children to an output stream.

Parameters

os	.
indent	.

The node is written with indentation according to its depth in the tree.

Parameters

os	Output stream to write to.
indent	Indentation level of the node.

Implements [XmlNode](#).

4.7.3.16 setName()

```
void XmlNode::setName (
    const String & name)
```

.

Parameters

name	.
------	---

The documentation for this class was generated from the following files:

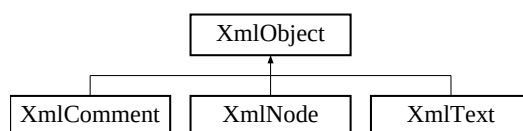
- XML/XmlNode.h
- XML/XmlNode.cpp

4.8 XmlObject Class Reference

XML .

```
#include <XmlObject.h>
```

Inheritance diagram for XmlObject:



Public Member Functions

- virtual void [serialize](#) (std::ostream &os, size_t indent=0) const =0
- virtual [XmlObject](#) * [clone](#) () const =0
- virtual ~XmlObject ()=default
- virtual void [printText](#) (std::ostream &os) const =0
- void [printIndents](#) (std::ostream &os, size_t indent=0) const

4.8.1 Detailed Description

XML .

, XML (,)

4.8.2 Member Function Documentation

4.8.2.1 clone()

virtual [XmlObject](#) * XmlObject::clone () const [pure virtual]

.

Returns

.

Implemented in [XmlComment](#), [XmlNode](#), and [XmlText](#).

4.8.2.2 printIndents()

void XmlObject::printIndents (
 std::ostream & os,
 size_t indent = 0) const

.

Parameters

os	.
indent	.

4.8.2.3 printText()

virtual void XmlObject::printText (
 std::ostream & os) const [pure virtual]

.

Parameters

os	.
----	---

Implemented in [XmlComment](#), [XmlNode](#), and [XmlText](#).

4.8.2.4 serialize()

```
virtual void XmlObject::serialize (
    std::ostream & os,
    size_t indent = 0) const    [pure virtual]
```

Parameters

os	.
indent	.

Implemented in [XmlComment](#), [XmlNode](#), and [XmlText](#).

The documentation for this class was generated from the following files:

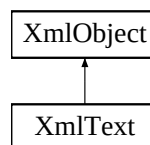
- XML/XmlObject.h
- XML/XmlObject.cpp

4.9 XmlText Class Reference

XML .

```
#include <XmlText.h>
```

Inheritance diagram for XmlText:



Public Member Functions

- const [String](#) & [getText](#) () const
- [XmlText](#) (const [String](#) &text)
- void [printText](#) (std::ostream &os=std::cout) const override
- void [serialize](#) (std::ostream &os, size_t indent=0) const override
- [XmlObject](#) * [clone](#) () const override

Public Member Functions inherited from [XmlObject](#)

- virtual `~XmlObject ()=default`
- void [printIndents](#) (std::ostream &os, size_t indent=0) const

4.9.1 Detailed Description

XML .

, XML , , .

4.9.2 Constructor & Destructor Documentation

4.9.2.1 XmlText()

XmlText::XmlText (
const [String](#) & text)

Parameters

text	.
------	---

4.9.3 Member Function Documentation

4.9.3.1 clone()

[XmlObject](#) * XmlText::clone () const [override], [virtual]

Returns

Implements [XmlObject](#).

4.9.3.2 getText()

const [String](#) & XmlText::getText () const

Returns

4.9.3.3 printText()

```
void XmlText::printText (  
    std::ostream & os = std::cout) const    [override], [virtual]
```

Parameters

os	.
----	---

Implements [XmlObject](#).

4.9.3.4 serialize()

```
void XmlText::serialize (  
    std::ostream & os,  
    size_t indent = 0) const    [override], [virtual]
```

Parameters

os	.
indent	.

Implements [XmlObject](#).

The documentation for this class was generated from the following files:

- XML/XmlText.h
- XML/XmlText.cpp

Chapter 5

File Documentation

5.1 CommandListener.h

```
00001 #pragma once
00002 #include "../XML/XmlFile.h"
00003 class CommandListener {
00004 public:
00005     void run(XmlFile& file);
00006 };
```

5.2 Vector.hpp

```
00001 #pragma once
00002 #include <iostream>
00003 template<typename T>
00004 class Vector {
00005 private:
00006     class Iterator {
00007     private:
00008         T* curr;
00009     public:
00010         Iterator(T* curr) : curr(curr) {}
00011         Iterator& operator++() {
00012             curr++;
00013             return *this;
00014         }
00015         bool operator!=(const Iterator& other) const {
00016             return curr != other.curr;
00017         }
00018         T& operator*() {
00019             return *curr;
00020         }
00021         const T& operator*() const {
00022             return *curr;
00023         }
00024     };
00025 public:
00026     Iterator begin() {
00027         return Iterator(arr);
00028     }
00029     Iterator end() {
00030         return Iterator(arr + size);
00031     }
00032     Iterator begin() const {
00033         return Iterator(arr);
00034     }
00035     Iterator end() const {
00036         return Iterator(arr + size);
00037     }
00038 public:
00039     bool empty() const {
00040         return size == 0;
00041     }
00042     operator bool() const {
```

```

00044     return !empty();
00045 }
00046 size_t getSize() const {
00047     return size;
00048 }
00049 void print(std::ostream& os = std::cout) const {
00050     for (size_t i = 0; i < size; i++)
00051     {
00052         os « arr[i] « ' ';
00053     }
00054 }
00055
00056 void push_back(const T& el) {
00057     if (size >= capacity)
00058     {
00059         resize();
00060     }
00061     arr[size] = el;
00062     size++;
00063 }
00064
00065 T pop_back() {
00066     size--;
00067     return arr[size];
00068 }
00069 T& operator[](size_t ind) {
00070     return arr[ind];
00071 }
00072 const T& operator[](size_t ind) const {
00073     return arr[ind];
00074 }
00075 void clear() {
00076     size = 0;
00077 }
00078
00079 private:
00080     T* arr;
00081     size_t size;
00082     size_t capacity;
00083
00084 private:
00085     void copy(const Vector& other) {
00086         size = other.size;
00087         capacity = other.capacity;
00088         arr = new T[capacity];
00089         for (size_t i = 0; i < size; i++)
00090         {
00091             arr[i] = other.arr[i];
00092         }
00093     }
00094     void moveFrom(Vector&& other) {
00095         size = other.size;
00096         capacity = other.capacity;
00097         arr = other.arr;
00098         other.arr = nullptr;
00099         other.capacity = 0;
00100         other.size = 0;
00101     }
00102     void resize() {
00103         if (capacity > 0)
00104         {
00105             capacity = capacity * 2;
00106         }
00107         else
00108         {
00109             capacity = 1;
00110         }
00111         T* newArr = new T[capacity];
00112         for (size_t i = 0; i < size; i++)
00113         {
00114             newArr[i] = arr[i];
00115         }
00116         delete[] arr;
00117         arr = newArr;
00118     }
00119     void free() {
00120         delete[] arr;
00121         arr = nullptr;
00122         size = 0;
00123         capacity = 0;
00124     }
00125 public:
00126     Vector() : arr(nullptr), size(0), capacity(0) {}
00127     Vector(Vector&& other) {
00128         moveFrom(std::move(other));
00129     }
00130     Vector(const Vector& other) {

```

```

00131     copy(other);
00132 }
00133 Vector& operator=(const Vector& other) {
00134     if (&other != this)
00135     {
00136         free();
00137         copy(other);
00138     }
00139     return *this;
00140 }
00141 Vector& operator=(Vector&& other) {
00142     if (&other != this)
00143     {
00144         free();
00145         moveFrom(std::move(other));
00146     }
00147     return *this;
00148 }
00149 ~Vector() {
00150     free();
00151 }
00152 };
00153 template<typename T>
00154 std::ostream& operator<<(std::ostream& os, const Vector<T>& v) {
00155     v.print(os);
00156     return os;
00157 }

```

5.3 String.h

```

00001 #pragma once
00002 #include <ostream>
00003 #include <cstddef>
00011 class String {
00012     char* str;
00013     size_t len;
00014     size_t capacity;
00015 private:
00017     void copy(const String& other);
00023     void free();
00028     void moveFrom(String&& other);
00034     void resizeTo(size_t newCapacity);
00040 public:
00042     String();
00046     String(const String& other);
00052     String(String&& other);
00058     String(const char* str);
00064     String& operator=(const String& other);
00071     String& operator=(String&& other);
00078     ~String();
00083     friend std::ostream& operator<<(std::ostream& os, const String& other);
00088     char& operator[](size_t ind);
00095     void append(const String& other);
00101     String& operator+=(const String& other);
00108     friend bool operator<(String const& lhs, String const& rhs);
00113     friend String operator+(const String& lhs, const String& rhs);
00119     friend std::istream& operator>>(std::istream& is, String& str);
00124     void clear();
00129     const char* getPtr() const;
00135     const char& operator[](size_t index) const;

```

```

00143
00148     operator bool() const;
00149
00153     friend bool operator==(String const& lhs, String const& rhs);
00154
00159     void push_back(char c);
00160
00165     size_t getLen() const;
00166 };

```

5.4 XmlAttribute.h

```

00001 #pragma once
00002 #include "../String/String.h"
00009 class XmlAttribute {
00010 private:
00011     String name;
00012     String value;
00013
00014 public:
00015
00019     XmlAttribute();
00020
00026     XmlAttribute(const String& name, const String& value);
00027
00032     void setValue(const String& val);
00033
00038     const String& getName() const;
00039
00044     const String& getValue() const;
00045 };

```

5.5 XmlComment.h

```

00001 #pragma once
00002 #include "../String/String.h"
00003 #include "XmlObject.h"
00010 class XmlComment : public XmlObject {
00011     String comment;
00012
00013 public:
00014
00019     const String& getComment() const;
00020
00026     void serialize(std::ostream& os, size_t indent = 0) const override;
00027
00032     void printText(std::ostream& os = std::cout) const override;
00033
00038     XmlObject* clone() const override;
00039
00044     XmlComment(const String& str);
00045 };

```

5.6 XmlFile.h

```

00001 #pragma once
00002 #include "XmlNode.h"
00003 #include "../String/String.h"
00004 #include "../Containers/Vector.hpp"
00005 #include "XmlComment.h"
00006 #include "XmlText.h"
00007 #include <map>
00015 class XmlFile {
00016     XmlNode* root;
00017     size_t nextId;
00018     std::map<String, XmlNode*> registry;
00019     String currentFileName;
00020
00021 private:
00022
00027     void copy(const XmlFile& other);
00028
00032     void free();
00033

```

```

00038     void moveFrom(XmlFile&& other);
00039
00044     void rebuildRegistry(XmlNode* curNode);
00045
00046 public:
00047
00051     XmlFile();
00052
00057     XmlFile(const XmlFile& other);
00058
00063     XmlFile(XmlFile&& other);
00064
00070     XmlFile& operator=(const XmlFile& other);
00071
00077     XmlFile& operator=(XmlFile&& other);
00078
00082     ~XmlFile();
00083
00088     void deserialize(std::istream& is);
00089
00094     void serialize(std::ostream& os) const;
00095
00101     XmlNode* getById(String id);
00102
00108     XmlComment* createComment(std::istream& is);
00109
00115     XmlNode* createNode(std::istream& is);
00116
00122     void addAttribute(std::istream& is, XmlNode& child);
00123
00129     String createValue(std::istream& is);
00130
00136     String createAttribute(std::istream& is);
00137
00144     void setName(std::istream& is, XmlNode& child, char first);
00145
00151     String toString(size_t num);
00152
00158     void select(const String& id, const String& key) const;
00159
00166     void set(const String& id, const String& key, const String& value);
00167
00172     void children(const String& id) const;
00173
00179     void child(const String& id, size_t pos) const;
00180
00185     void text(const String& id) const;
00186
00192     void deleteAttr(const String& id, const String& key);
00193
00198     void newChild(const String& id);
00199
00204     void open(const String& filePath);
00205
00209     void close();
00210
00214     void save() const;
00215
00220     void saveAs(const String& fileName) const;
00221
00227     const XmlNode& getByTagName(const String& tagName) const;
00228
00235     XmlText* createText(std::istream& is, char first);
00236
00241     void reachEndOfTag(std::istream& is);
00242
00248     bool isWhitespaceOnly(const String& str);
00249
00255     void buildTree(std::istream& is, XmlNode& curNode);
00256
00261     void handleId(XmlNode& child);
00262
00268     const XmlNode* getById(const String& id) const;
00269
00273     Vector<XmlNode*> applyQuery(const Vector<XmlNode*>& curResult, const String& miniQuery) const;
00274
00278     Vector<String> splitBySlash(const String& query) const;
00279
00283     Vector<XmlNode*> filterInd(const Vector<XmlNode*>& nodes, const String& miniQuery, size_t filterStart) const;
00284
00288     Vector<XmlNode*> filterAllAttr(const Vector<XmlNode*>& nodes, const String& miniQuery, size_t filterStart) const;
00289
00293     Vector<XmlNode*> filterByText(const Vector<XmlNode*>& nodes, const String& miniQuery, size_t filterStart) const;
00294
00298     String getNameFromQuery(const String& query, size_t start = 0) const;
00299
00303     Vector<XmlNode*> applyFilter(const Vector<XmlNode*>& nodes, const String& miniQuery, size_t filterStart) const;

```

```

00304
00309 void xpath(const String& query) const;
00310 };

```

5.7 XmlNode.h

```

00001 #pragma once
00002 #include "XmlObject.h"
00003 #include "XmlAttribute.h"
00004 #include "../Containers/Vector.hpp"
00005 #include "../String/String.h"
00013 class XmlNode : public XmlObject {
00014 private:
00015     String name;
00016     Vector<XmlAttribute> attributes;
00017     Vector<XmlObject*> children;
00018     String id;
00019
00020 private:
00021
00026     void copy(const XmlNode& other);
00027
00031     void free();
00032
00037     void moveFrom(XmlNode&& other);
00038
00039 public:
00040
00044     XmlNode();
00045
00050     XmlNode(const XmlNode& other);
00051
00056     XmlNode(XmlNode&& other);
00057
00063     XmlNode& operator=(XmlNode&& other);
00064
00070     XmlNode& operator=(const XmlNode& other);
00071
00075     ~XmlNode();
00076
00081     const String& getId() const;
00082
00087     void addAttribute(const XmlAttribute& attribute);
00088
00093     void addChild(XmlObject* child);
00094
00099     const Vector<XmlAttribute>& getAttributes() const;
00100
00105     Vector<XmlAttribute>& getAttributes();
00106
00111     void printText(std::ostream& os = std::cout) const override;
00112
00117     String getTextContent() const;
00118
00124     Vector<XmlNode*> getChildrenByName(const String& name) const;
00125
00131     int getAttributeInd(const String& attr) const;
00132
00137     XmlNode(const String& name);
00138
00143     const String& getName() const;
00144
00149     void setName(const String& name);
00150
00155     const Vector<XmlObject*>& getChildren() const;
00156
00162     void serialize(std::ostream& os, size_t indent = 0) const override;
00163
00168     XmlObject* clone() const override;
00169 };

```

5.8 XmlObject.h

```

00001 #pragma once
00002 #include <iostream>
00010 class XmlObject {
00011 public:
00012
00018     virtual void serialize(std::ostream& os, size_t indent = 0) const = 0;

```

```
00019
00024     virtual XmlObject* clone() const = 0;
00025
00029     virtual ~XmlObject() = default;
00030
00035     virtual void printText(std::ostream& os) const = 0;
00036
00042     void printIndents(std::ostream& os, size_t indent = 0) const;
00043 };
```

5.9 XmlText.h

```
00001 #pragma once
00002 #include "XmlObject.h"
00003 #include "../String/String.h"
00011 class XmlText : public XmlObject {
00012     String text;
00013
00014 public:
00015
00020     const String& getText() const;
00021
00026     XmlText(const String& text);
00027
00032     void printText(std::ostream& os = std::cout) const override;
00033
00039     void serialize(std::ostream& os, size_t indent = 0) const override;
00040
00045     XmlObject* clone() const override;
00046 };
```


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